Implicit function thm in Banach spaces:

Let T, X, Y be Banach spaces

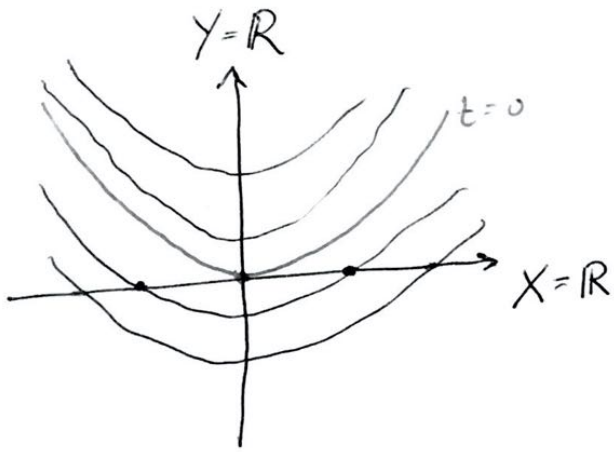
$F : T \times X \longrightarrow Y$ smooth (or just C^1)

s.t.
- $F(0,0) = 0$
- $D_2F(0,0) : X \to Y$ is surjective with finite dim kernel

$\Longrightarrow$ $\exists$ neighborhood $0 \in \mathcal{U} \subset T$ and $x : \mathcal{U} \to X$ continuous

s.t. $x(0) = 0$

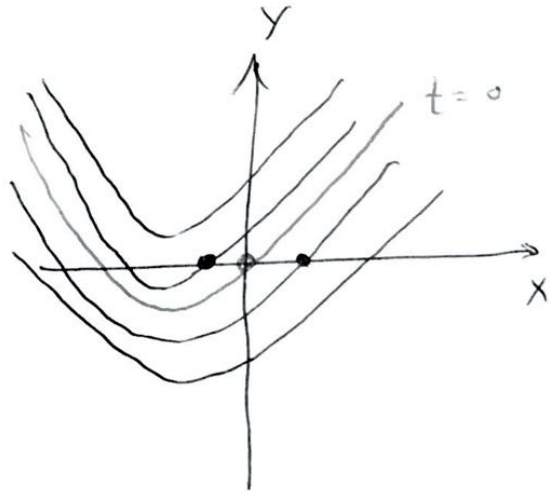
and $F(t, x(t)) = 0$ for $t \in \mathcal{U}$



$F(t,x) = x^2 + t$

$D_2F(0,0) = 0$

$(\mathbb{R} \to \mathbb{R})$



$F(t,x) = x^2 + 2x + t$

$D_2F(0,0) = 2$

Need to show that f_t are smooth, given that $f_t \in C^{3,\alpha}$.

In local coords, f_t satisfies

$$\log \frac{\det(g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f_t)}{\det(g_{j\bar{k}})} = tF + c_t$$

take ∂_ℓ derivative:

$$\tilde{g}^{j\bar{k}} \partial_\ell (g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f_t) - g^{j\bar{k}} \partial_\ell g_{j\bar{k}} = t \partial_\ell F$$

$$\tilde{g}_{j\bar{k}} = g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f \leftarrow C^{1,\alpha}$$

write as

$$C^{1,\alpha} \longrightarrow \tilde{g}^{j\bar{k}} \partial_j \partial_{\bar{k}} (\partial_\ell f_t) = \underbrace{t \partial_\ell F + g^{j\bar{k}} \partial_\ell g_{j\bar{k}} - \tilde{g}^{j\bar{k}} \partial_\ell g_{j\bar{k}}}_{\in C^{1,\alpha}} \quad (*)$$

Schauder regularity $\Longrightarrow$ $\partial_\ell f_t \in C^{3,\alpha}$

$\Longrightarrow$ $f_t \in C^{4,\alpha}$

$(*)$ $\Longrightarrow$ $\partial_\ell f_t \in C^{4,\alpha} \Longrightarrow f_t \in C^{5,\alpha}$

$\ldots \Longrightarrow f_t \in C^{k,\alpha} \ \forall k$

so f_t is smooth

(Bootstrapping) $\square$

Closedness

Theorem (A priori estimates)

(M, g) compact Kähler, $F \in C^\infty(M)$, $f : M \to \mathbb{R}$

satisfies
$$\left.\begin{aligned} \det\left(g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f\right) &= e^F \det\left(g_{j\bar{k}}\right) \\ g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f &> 0 \\ \sup_M f &= 0 \end{aligned}\right\} \quad (*)$$

Then $\|f\|_{C^{3,\alpha}(M)} < C$ for C depending on (M, g), $\|F\|_{C^3(M)}$.

Step 1 Bounding $\|f\|_{L^1}$, i.e. $\int_M f \, dV_g$ from below

Only uses conditions $g_{j\bar{k}} + \partial_j \partial_{\bar{k}} f > 0$, $\sup_M f = 0$.

Preliminaries 1. Can cover M with finitely many charts U_i s.t. g admits local Kähler potentials $\varphi_i : U_i \to \mathbb{R}$

s.t. $g_{j\bar{k}} = \partial_j \partial_{\bar{k}} \varphi_i$ (local $\partial\bar{\partial}$-lemma)

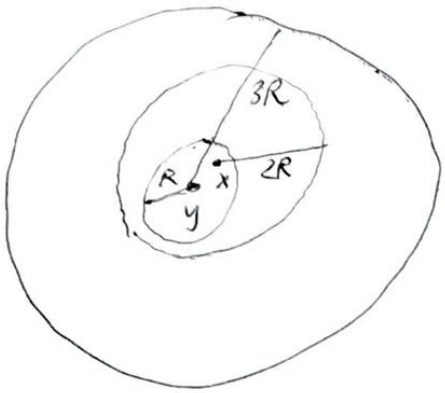
$\varphi_i \leq 0$

then $\partial_j \partial_{\bar{k}} (\varphi_i + f) > 0$

so $\Delta^{Euc} (\varphi_i + f) > 0$ on chart

2. mean value inequality :

if $\Delta u \geq 0$, $u \leq 0$ on a ball $B(y, 3R)$



and $x \in B(y, R)$ then

$$u(x) \leq \frac{1}{\mathrm{vol}(B(x, 2R))} \int_{B(x,2R)} u$$

$$\leq \frac{1}{\mathrm{vol}(B(x, 2R))} \int_{B(y,R)} u$$

$$\Longrightarrow \quad \|u\|_{L^1(B(y,R))} \leq \mathrm{vol}(B(x,2R))\, |u(x)|$$

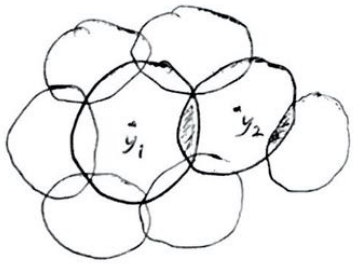
<u>Estimate $\|f\|_{L^1}$ as follows</u> : Cover M with balls $B(y_i, 3R_i)$, each contained in a coord. chart, s.t. $B(y_i, R_i)$ still cover M.

Let $p \in M$ s.t. $f(p) = \sup_M f = 0$

assume $p \in B(y_1, R_1)$

Apply mean value ineq. to $\varphi_1 + f$

$$\Longrightarrow \quad \|f\|_{L^1(B(y_1, R_1))} \leq C_1 \quad \left(\begin{array}{c}\text{dep. only} \\ \text{on } (M, g)\end{array}\right)$$



we get $\sup_{B(y_i, R_i)} f \geq -C_2$

whenever $B(y_i, R_i) \cap B(y_1, R_1) \neq \emptyset$

mean value ineq $\Longrightarrow$ $\|f\|_{L^1(B(y_i, R_i))} \leq C_3$

repeat this argument and after finitely many steps we cover all of M

so $\|f\|_{L^1(M)} \leq C$ $\square$